

ECON 5345 Homework 2 Report

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Question 1

a. Q1

Question 2

- a. The impulse response function is shown in Figure 1.

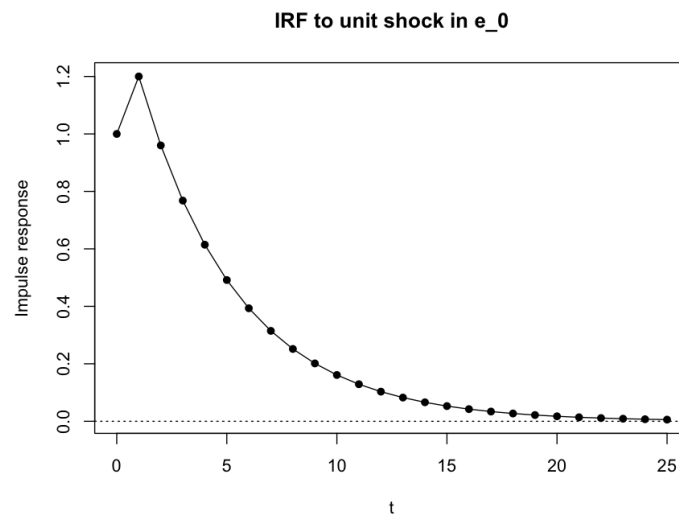


Figure 1: IRF to unit shock in e_0

- b.

Question 3

a. Q3

Question 4

a. The expectation is

$$\begin{aligned}\mathbb{E}[x_t] &= \mathbb{E}[\alpha] \cos t + \mathbb{E}[\beta] \sin t \\ &= 0.\end{aligned}$$

The variance is

$$\begin{aligned}\text{Var}[x_t] &= \text{Var}[\alpha] \cos^2 t + \text{Var}[\beta] \sin^2 t + 2 \text{Cov}(\alpha, \beta) \cos t \sin t \\ &= \cos^2 t + \sin^2 t \\ &= 1.\end{aligned}$$

The second line is by the independence of α and β .

b. No. Note that

$$\begin{aligned}\mathbb{E}[\alpha^2] &= \text{Var}[\alpha] + (\mathbb{E}[\alpha])^2 \\ &= 1 \\ \mathbb{E}[\beta^2] &= \text{Var}[\beta] + (\mathbb{E}[\beta])^2 \\ &= 1 \\ \mathbb{E}[\alpha\beta] &= \text{Cov}(\alpha, \beta) + \mathbb{E}[\alpha]\mathbb{E}[\beta] \\ &= 0.\end{aligned}$$

As a result, we have

$$\begin{aligned}\mathbb{E}[x_t x_{t-k}] &= \mathbb{E}[\alpha^2] \cos t \cos(t-k) + \mathbb{E}[\beta^2] \sin t \sin(t-k) \\ &\quad + \mathbb{E}[\alpha\beta][\cos t \sin(t-k) + \sin t \cos(t-k)] \\ &= \cos t \cos(t-k) + \sin t \sin(t-k) \\ &= \cos[t - (t-k)] \\ &= \cos k.\end{aligned}$$